

Sliding Puzzle

How the exhibit reaches schools, museums, and the public — during and after the project

Live exhibit: <https://erikaroldanroa.github.io/ai-exhibits-rl-public/> · EN · FR · DE · IT

Source code: github.com/ErikaRoldanRoa/ai-exhibits-rl-public

The asset

“Sliding Puzzle” is a browser-native exhibit in which visitors train their own Q-learning agent on sliding puzzles of growing size (2×2, 2×3, 3×3) and experience the curse of dimensionality firsthand. It runs in four languages, requires no installation or account, works offline after the first visit, and ships with printable A4 classroom kits and full teacher-facing documentation (handbook entries in EN · FR · DE · IT). Because the exhibit is a static web application, it has no server costs: it remains permanently available at zero marginal cost — the structural basis of this exploitation plan.

Evidence from pilots

Institut Henri Poincaré — Maison Poincaré, Paris — prototype workshop, 30 January 2026. Two Seconde school groups met the Sliding Puzzles RL exhibit across morning and afternoon sessions (n = 51). Interest ran high: a mean of 3.35 / 4 on an adapted LES instrument (Dietiker et al., 2019), with 92% of students reporting themselves interested or very interested, and a large effect against the literature benchmark ($p < .001$, $d = 0.88$). When students were invited to describe the experience in their own terms, 138 of their 153 descriptor selections (90.2%) were positive — the analytics confirming what the room felt like. The filled questionnaires are available in the AI-Exhibits shared Drive.

University of Geneva — girls’ mathematics camp, 13 April 2026. The exhibit anchored a lab session with 29 participants, run entirely from the public hub with the printed team scoresheets. Earlier deployments include a bilingual DE+FR workshop (February 2026) and a French print-kit deployment for Pi Day, Paris (March 2026).

Exploitation channels

- **Permanent public hub.** The exhibit stays online after the project ends, free, in four languages, at the address above — usable tomorrow by any teacher with a browser, with no accounts and no data collection by default.
- **Open repository.** Source, documentation, and printable kits are published on GitHub, with the handbook entries (EN · FR · DE · IT) archived in the repository’s [documentation folder](#).
- **Workshops and events.** A request-an-event channel opens in July 2026: time-bounded workshops receive auto-expiring credentials; permanent installations (museums, paper-linked tournaments) receive open credentials. The 75-minute facilitator session plan and kits make delivery transferable to non-specialist facilitators.

- **E-Guide.** The exhibit is Activity 1.7 of the partnership's E-Guide (WP4), giving educators a guided pathway into the same materials.
- **Research dissemination.** A pedagogical paper seeded by this exhibit is in preparation for DMVM (submission June 2026), and the three research papers behind the mathematics (arXiv:2412.01937, 2307.14143, 2201.00919) link back to the exhibit.
- **Institutional continuation.** MPI MiS (Leipzig) continues to host AI workshops built on the exhibit; materials and an online induction have been offered so that other MPI scientists can deliver it independently, including at public-facing events (Tag der offenen Tür, Lange Nacht der Wissenschaften).

Sustainability and access

License: CC BY-NC-ND 4.0 — free non-commercial sharing with attribution; adaptations and translations by written permission, which keeps quality control while allowing growth. Feedback flows through the in-app ⓘ channel (open to all, no account); community contributions will route through a structured form from July 2026. The exhibit was audited for accessibility, French typography, internationalisation, and mathematical accuracy; God's Numbers are verified by a reproducible breadth-first-search script published in the project repository.



Co-funded by the European Union

Prepared by Dr. Erika Roldán Roa (Max Planck Institute for Mathematics in the Sciences, Leipzig) for the AI-Exhibits Erasmus+ partnership: MPI MiS · IMAGINARY · Institut Henri Poincaré · Citizens in Power · ITT Giordani Striano.